

controls the functioning activities of each and every part of our body, but the brain waves can also be used to affect living and non-living objects through telepathy, telekinesis, hypnotism, intuition, love at first sight, etc. Now, it is becoming possible to operate advanced and smart electronic devices with brain waves.

Matter waves

Wave function was supposed to be a characteristic of particles at quantum level, but in 1924, Louis de Broglie (Nobel Prize in Physics in 1929) proposed that a wave function is associated with all particles. The standard interpretation from the intensity of a wave function of a particle at any point is the probability of finding the particle at that point. The wavelengths of the harmonic waves are used to build the wave function called a matter wave.

Thus, matter waves are a central part of the theory of quantum mechanics leading to the wave-particle duality hypothesis. According to that, all matter can exhibit wave-like behavior. For example, a beam of electrons can be diffracted just like a beam of light or a water wave. A wave (mechanical or electromagnetic) can be described as a disturbance that transports energy from one location (its source) to another location, with or without a medium, but without transporting matter.

Electromagnetic (EM) waves are generated as a result of vibrations between an oscillating electric field and a magnetic field. The entire EM spectrum includes waves representing from the lowest to the highest frequency (or longest to shortest wavelength), that is, all radio waves, microwaves, infrared radiation, visible light, ultraviolet radiation, X-rays and gamma rays. However, matter waves of very low frequency (beyond EM spectrum) do exist but are difficult to detect and differentiate. All waves are made up of photons that travel through space and interact with matter, giving rise to various phenomena like absorption, reflection, interference and diffraction.

Brain waves

The human brain emits electrical waves, which produce different effects on the body. Practically, the distinc-

Brain Waves Names

Wave	Frequency range	State of mind and body
Infra-Low	<0.5Hz	Slow cortical potentials. Very little is known about infra-low brain-waves because, due to their slow nature, these are difficult to detect and accurately measure
Delta	0.5Hz–4Hz	Deep sleep. Healing and regeneration are stimulated in this state
Theta	4Hz–8Hz	Drowsiness (also first stage of sleep). Theta is our gateway to learning, memory and intuition
Alpha	8Hz–14Hz	These waves represent the overall coordination, calmness, alertness of mental state, mind and body integration, and learning
Beta	14Hz–30Hz	Highly alert and focused
Gamma	38Hz-42Hz	Fastest brain activity responsible for cognitive functioning, learning, memory, and information processing. It is considered that gamma brain wave rhythms are responsible for modulated perception, consciousness and spiritual emergence

tion between the brain waves is difficult due to the presence of different frequencies. There are four brainwave states common to the human species, which range from the high-amplitude, low-frequency delta to the low-amplitude, high-frequency beta. These brain wave ranges specify different states of human mind, from deep sleep to high arousal. Neuroscientists have named these waves as shown in the table.

Electronics of brain waves

The electrical oscillations called brain waves have intrigued scientists and the public for a long time, and their function is still debated. The propagation of brain waves within brain as well as outside brain, and their (propagation) effects are under investigation. Scientists and engineers are putting their sincere efforts for developing and designing suitable instrumentation for the detection and monitoring of brain waves.

The neurons in the brain transmit signals and a voltage is formed that the electrodes detect and transmit onwards through a tiny amplifier. Electroencephalography (EEG) is a well known and established electrophysiological monitoring method to record electrical activity of the brain. EEG is a non-invasive method with the electrodes/sensors placed along the scalp. However, invasive electrodes are also being used in what is known as electrocorticography technique of brain monitoring.

Advances are being made towards the development of invasive technology of brain recording as well. In the times to come, seamless and

minimally invasive three-dimensional interpretation of electronics of brain could allow for continuous monitoring and manipulation of brain waves and their properties and effects. Thanks to non-invasive tools that have been around for decades, such as EEG and functional magnetic resonance imaging (fMRI), physicians and neuroscientists can measure changes in the brain without drilling a hole in the skull. Thus, the next technology to pervade our lives may be the so-called neural interfaces.

Understanding brain waves coupling with electronic components is crucial not only to collect information about brain signaling but also to diagnose and treat neurological disorders and diseases. Human tissue being elastic and mobile, damage and inflammation can arise at the interface with rigid electronic components. Therefore flexible electronics is needed to not only provide useful means for conforming electronics free from such problems but also take it to non-planar surfaces.

For long-term stable neural recording, scientists have developed a new conductive bio-compatible material that is as soft as human tissue and can be stretched to twice its length. The material consists of gold-coated titanium dioxide nanowires, embedded into silicone rubber.

Use of electronic implants is progressing and scientists have developed methods to inject microelectronic devices such as wires and transistors directly into the brain to measure or stimulate neural activity. Neuro-philic electronic sensors are one million